

# HOUSE PRICE PREDICTION USING LINEAR REGRESSION

## Artificial Intelligence & Machine Learning – Task 1

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### 1. Introduction

This project focuses on predicting house prices using the California Housing Dataset. A Linear Regression model was developed to understand the relationship between housing features and house values. The project demonstrates the complete machine learning workflow including data loading, exploratory data analysis, model training, evaluation, and visualization.

### 2. Dataset Description

The California Housing Dataset contains information about housing districts such as median income, house age, average rooms, population, occupancy, latitude, and longitude. The target variable is Median House Value (MedHouseVal).

### 3. Exploratory Data Analysis (EDA)

EDA was performed to understand feature distributions and relationships. Histograms were generated for all variables to identify trends, skewness, and outliers.

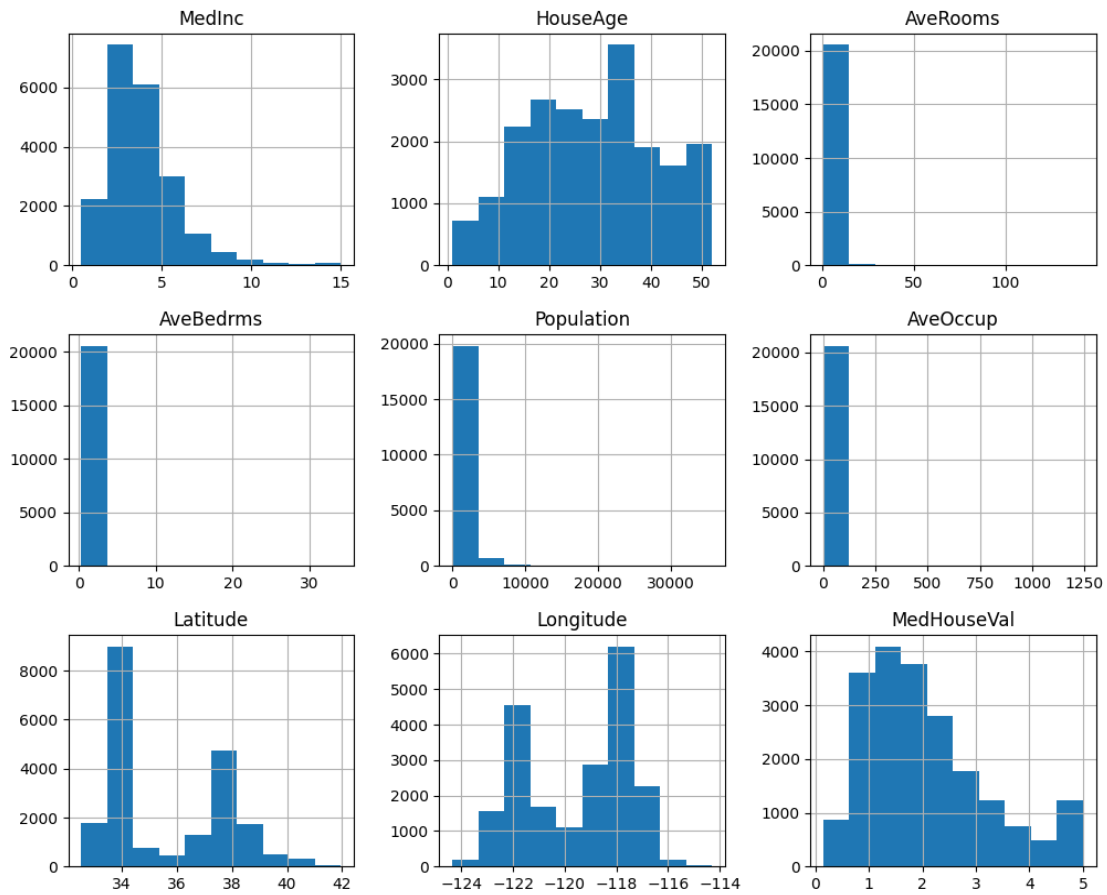


Figure 1: Histogram analysis of all dataset features.

## 4. Correlation Analysis

A correlation heatmap was generated to understand relationships among features. Median Income (MedInc) showed the strongest positive correlation with house prices (~0.69), indicating income is an important factor affecting house values.

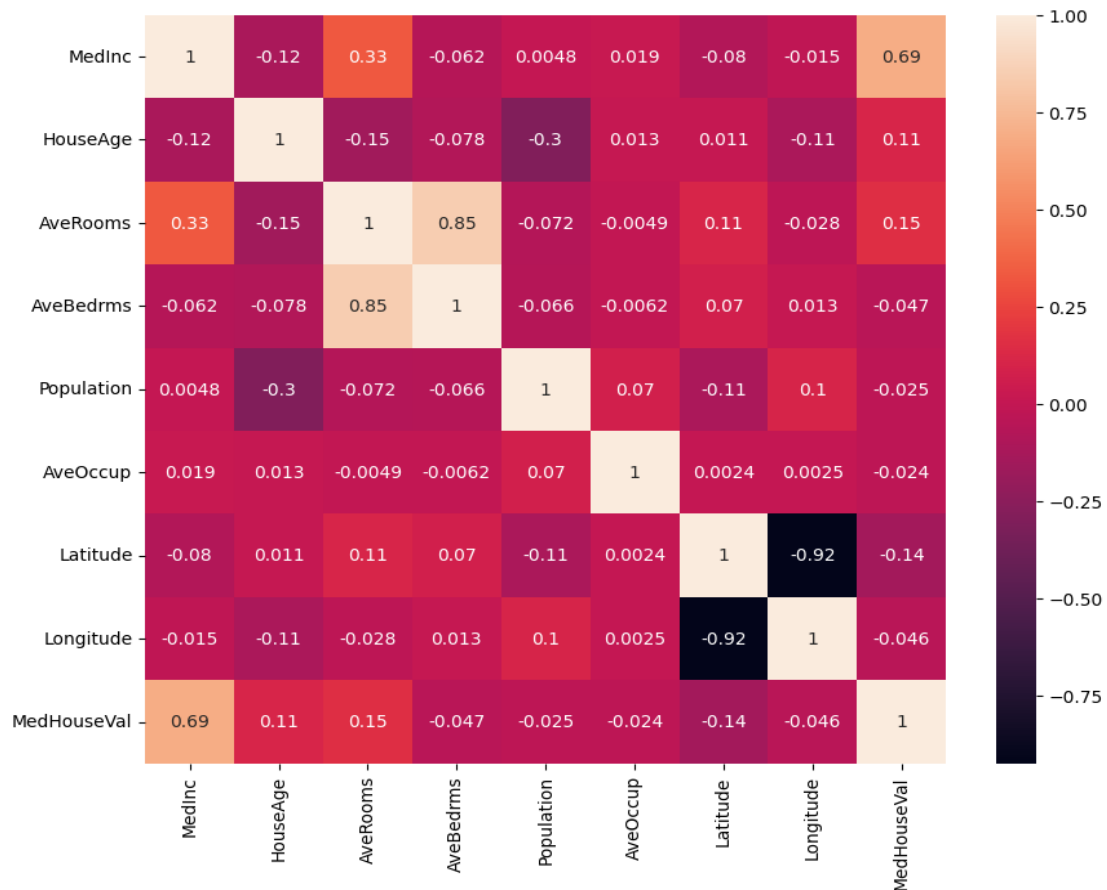


Figure 2: Correlation Heatmap.

## 5. Model Development and Evaluation

The dataset was split into training (80%) and testing (20%) sets. A Linear Regression model was trained using the training data and predictions were generated on the test dataset. Performance was evaluated using MAE, RMSE, and  $R^2$  Score.

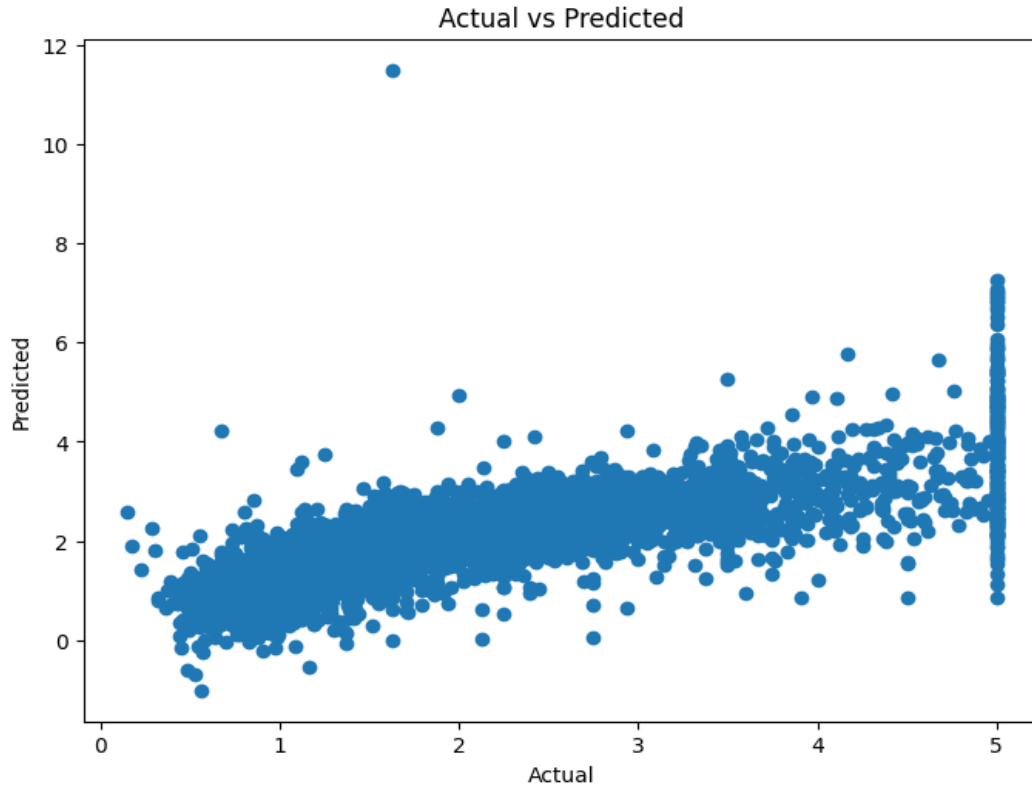


Figure 3: Actual vs Predicted House Prices.

## 6. Results

The scatter plot shows a positive relationship between actual and predicted values, indicating that the model successfully captured important trends in the data. Although some prediction errors exist, the model provides a reasonable baseline for house price prediction.

## 7. Conclusion

This project successfully implemented a Linear Regression model for house price prediction. The project covered data preprocessing, visualization, model training, and evaluation. Future improvements may include feature engineering, Random Forest Regression, Gradient Boosting, and XGBoost models.